

Script started on 2026-02-05 13:51:48-06:00 [TERM="xterm" TTY="/dev/pts/41" COLUMNS=80]
rc47379@ares:~\$ pwd

/home/students/rc47379
rc47379@ares:~\$ cat timer.info

NAME: Calvin Rice

CLASS: CSC122-002

TITLE: What runs all day but has no legs

LEVEL: 3

DESCRIPTION:

Timer class that counts elapsed time since its start functionrc47379@ares:~\$ show-c

timer.cpp:

```
1  #include <iostream>
2  #include <chrono>
3  #include <ctime>
4  #include <format>
5  #include <iomanip>
6
7  using namespace std;
8  using namespace chrono;
9
10 class Timer
11 {
12 private:
13     bool running;
14     time_point<system_clock> startTime;
15 public:
16     int elapsedTime;
17     Timer()
18     {
19         elapsedTime = 0;
20         running = false;
21     }
22
23     void start()
24     {
25         running = true;
26         startTime = system_clock::now();
27         elapsedTime = 0;
28     }
29
30     void stop()
31     {
```

```
32         running = false;
33         elapsedTime = duration_cast<seconds>(system_clock::now() - startTi
34     }
35
36     int elapsed()
37     {
38         if(running)
39         {
40             elapsedTime = duration_cast<seconds>(system_clock::now() - sta
41         }
42         return elapsedTime;
43     }
44
45     void output()
46     {
47         cout << elapsed();
48     }
49
50     Timer add(Timer _t)
51     {
52         Timer sum;
53         sum.elapsedTime = elapsed() + _t.elapsed();
54         return sum;
55     }
56
57     Timer subtract(Timer _t)
58     {
59         Timer diff;
60         diff.elapsedTime = elapsed() - _t.elapsed();
61         return diff;
62     }
63 };
64
65 int main()
66 {
67     Timer t;
68     bool repeat = true;
69     string input;
70
71     while (repeat)
72     {
73         time_t now = time(0);
74         tm local_tm{};
75         //localtime_r(&now, &local_tm);
76         cout << "Current Time: " << std::ctime(&now) << endl;
77
78         t.start();
79
80         cout << "Press Enter to stop the timer" << endl;
81         cin.get();
82
83         t.stop();
84
85         cout << "Elapsed Time: " << t.elapsed() << endl;
```

```

86
87     now = time(0);
88     //localtime_r(&now, &local_tm);
89     cout << "Current Time: " << ctime(&now) << endl;
90
91     bool asking = true;
92     while(asking)
93     {
94         cout << "Repeat? Enter y/n" << endl;
95         cin >> input;
96         if(input[0] == 'y')
97         {
98             cout << "Repeating..." << endl;
99             asking = false;
100        }
101        else if(input[0] == 'n')
102        {
103            return 0;
104        }
105        else
106        {
107            cout << "Invalid input" << endl;
108        }
109    }
110 }
111
112 return 0;
113 }

```

rc47379@ares:~\$ CPP timer

timer.cpp***

timer.cpp: In constructor

'Timer::Timer()':

timer.cpp:17:5: warning:

'Timer::running' should be initialized in the member initialization list [-Weffc++]

```

17 |     Timer()
    |     ^~~~~

```

timer.cpp:17:5: warning:

'Timer::startTime' should be initialized in the member initialization list [-Weffc++]

timer.cpp:17:5: warning:

'Timer::elapsedTime' should be initialized in the member initialization list [-Weffc++]

timer.cpp: In member function 'void

Timer::stop()':

timer.cpp:33:84: warning: conversion

from 'std::chrono::duration<long int>::rep' {aka 'long int'} to 'int' may change

value [-Wconversion]

```

33 |         elapsedTime =
    |         duration_cast<seconds>(system_clock::now() -
    |         startTime).count();
    |

```

timer.cpp: In member function 'int

Timer::elapsed()':

timer.cpp:40:88: warning: conversion

from 'std::chrono::duration<long int>::rep' {aka 'long int'} to 'int' may change

value [-Wconversion]

```

40 |         elapsedTime =
    |         duration_cast<seconds>(system_clock::now() -
    |         startTime).count();
    |

```

timer.cpp: In function 'int

main()':

timer.cpp:74:12: warning:

unused variable 'local_tm'

[-Wunused-variable]

```

74 |         tm local_tm{};
    |         ^~~~~~

```

rc47379@ares:~\$./timer.out

Current Time: Thu Feb 5 13:52:13 2026

Press Enter to stop the timer

Elapsed Time: 3

Current Time: Thu Feb 5 13:52:16 2026

Repeat? Enter y/n

y

Repeating...

Current Time: Thu Feb 5 13:52:18 2026

Press Enter to stop the timer

Elapsed Time: 0

Current Time: Thu Feb 5 13:52:18 2026

Repeat? Enter y/n

y

Repeating...

Current Time: Thu Feb 5 13:52:20 2026

Press Enter to stop the timer

Elapsed Time: 0

Current Time: Thu Feb 5 13:52:20 2026

Repeat? Enter y/n

y

```
Repeating...
Current Time: Thu Feb  5 13:52:30 2026

Press Enter to stop the timer
Elapsed Time: 0
Current Time: Thu Feb  5 13:52:30 2026

Repeat? Enter y/n
n
rc47379@ares:~$ cat timer.tpq

How many data members do you have? What data types are they? Where is this data type?
-3: boolean for if the timer is running, integer for seconds elapsed, and a ctime_t for the current time

How many constructors do you have? What is the purpose of each (i.e. in what situation would you use each)?
-Only 1 because the timer would start reset and not running

Is it okay that the mutators for the run state of the Timer are not named in the state?
-Yes because they are only used internally

Is it okay that the accessor for the elapsed seconds of the Timer is not named in the state?
-Yes because it doesn't matter

Does your output() method print anything besides the elapsed seconds (even an endl)?
-No because the user might want different text surrounding the output

Does your add() method change the value of the calling object? Should it? (Hint: Does it return a new object?)
-No, it returns a new object and doesn't change the 2 inputs

Why is an input method not a good/reasonable idea for the Timer class?
-There is nothing to input

Why don't you have a mutator for the elapsed time?
-The elapsed time wouldn't need to be changed because it is the difference between the current time and the start time

The proposed driver/test 'app' doesn't really test all of your class' functionality
-The driver doesn't use the add or subtract functions
rc47379@ares:~$ exit

exit

Script done on 2026-02-05 13:52:45-06:00 [COMMAND_EXIT_CODE="0"]
```